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I2S (BCLK J6-2 / IO46, DOUT J6-3 / IO45, WS J6-9 / IO3) -> PCM5102A (BCK PLL, no MCLK) -> TPA6138A2 (gain −1, ground-centred)
-> 3.5 mm jack (tip = left = HP_L, ring = right = HP_R, sleeve = GND). 100k pull-downs on the three lines, no pull-ups.
Power: P+ (raw cell) J6-12) -> TPS7A2033 3.3 V LDO, switched by EN; DET = charge pump on WS (R1 C3 D1 C4 R2): idle ~0.1 uA.
The amplifier is held in mute by R3/C5 for ~100 ms after the LDO turns on. J6 3V3 and I2C are not used.
Headers (pads named by the J6 pin they mate): J1 1x4 = pins 1–4 (GND, BCLK, DOUT, GND); J2 = pin 9 (WS); J3 = pin 12 (VBAT).
No pad and no hole at J6 5, 6, 7, 8, 10, 11. Jack pads: 1 sleeve, 2 ring, 3 tip (each on both tabs).

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PCM5102A I2S DAC + TPA6138A2 headphone driver + 3.5 mm jack; plugs into main-board J6
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Sheet: /
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Title: Silkscreen audio attachment

Size: A3	Date: 2026-11-09	**Rev: A**
KiCad E.D.A. 9.0.6		Id: 1/1